PDF → Markdown Converter

Ultimate Stress-Test Document

This document is **intentionally complex**. It contains *italic prose*, **bold headings**, ***bold-italic combinations***, nested tables, multi-column data, inline code, fenced code blocks in multiple languages, checklists (checked & unchecked), and edge-case formatting designed to break naive converters.

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1. Checklists & Task Tracking

The following lists test whether the converter correctly handles **checked**, *unchecked*, and ***indented*** checklist items. Items marked with **bold labels** are blocking issues; items in *italic* are optional.

1.1 Pre-flight Checklist

- [x] Repository cloned and branch checked out
- [x] Virtual environment activated (Python 3.11+)
- [x] Dependencies installed via pip install -r requirements.txt
- [ ] Environment variables set in .env file
- [ ] Database migrations applied: python manage.py migrate
- [ ] Secrets rotated in production vault
- [x] Linter passes: ruff check . --select ALL
- [ ] Type-checker passes: mypy --strict src/
- [ ] All unit tests passing: pytest -x -q
- [x] Docker image builds without warnings
- [ ] Integration tests passing against staging DB
- [ ] Performance benchmarks within 10% of baseline

1.2 Security Audit Checklist

- [x] OWASP Top 10 reviewed and mitigated
- [x] SQL injection: all queries use parameterised statements
- [ ] XSS: output encoding enforced at template layer
- [ ] CSRF tokens present on all state-changing forms
- [x] Authentication uses bcrypt with work factor >= 12
- [ ] Authorisation checks present on every API endpoint
- [ ] Sensitive data encrypted at rest (AES-256-GCM)
- [x] TLS 1.2+ enforced; TLS 1.0/1.1 disabled
- [ ] Dependency audit: no known CVEs in pip-audit report
- [ ] Rate limiting configured on login and signup routes
- [ ] Error messages do not leak stack traces to clients
- [x] Security headers set (CSP, HSTS, X-Frame-Options)
- [ ] Penetration test scheduled with external vendor

2. Code Blocks & Inline Code

Each snippet below uses a **different programming language** to stress-test syntax highlighting and indentation preservation. Pay special attention to lines with *mixed indentation*, long lines that may wrap, and **special characters** such as backslashes, angle brackets, and Unicode literals.

2.1 Python – Async HTTP Client

```python
# Python 3.11  |  async/await + type hints
from __future__ import annotations
import asyncio
import httpx
from typing import Any

BASE_URL = "https://api.example.com/v2"

async def fetch_paginated(
    endpoint: str,
    *,
    page_size: int = 100,
    headers: dict[str, str] | None = None,
) -> list[dict[str, Any]]:
    """Fetch all pages from a paginated REST endpoint."""
    results: list[dict[str, Any]] = []
    params = {"limit": page_size, "offset": 0}

    async with httpx.AsyncClient(base_url=BASE_URL, timeout=30.0) as client:
        while True:
            resp = await client.get(endpoint, params=params, headers=headers or {})
            resp.raise_for_status()
            data = resp.json()
            results.extend(data["items"])
            if len(data["items"]) < page_size:
                break
            params["offset"] += page_size

    return results


if __name__ == "__main__":
    items = asyncio.run(fetch_paginated("/products", page_size=50))
    print(f"Fetched {len(items)} items")
```

2.2 TypeScript – Generic Repository Pattern

```typescript
// TypeScript 5.x
interface Entity {
  id: string;
  createdAt: Date;
  updatedAt: Date;
}

interface Repository<T extends Entity> {
```

```
  findById(id: string): Promise<T | null>;
  findAll(filter?: Partial<T>): Promise<T[]>;
  save(entity: Omit<T, keyof Entity>): Promise<T>;
  delete(id: string): Promise<boolean>;
}

class InMemoryRepository<T extends Entity> implements Repository<T> {
  private store = new Map<string, T>();

  async findById(id: string): Promise<T | null> {
    return this.store.get(id) ?? null;
  }

  async findAll(filter?: Partial<T>): Promise<T[]> {
    const all = [...this.store.values()];
    if (!filter) return all;
    return all.filter(item =>
      Object.entries(filter).every(([k, v]) => (item as any)[k] === v)
    );
  }

  async save(data: Omit<T, keyof Entity>): Promise<T> {
    const entity = {
      ...data,
      id: crypto.randomUUID(),
      createdAt: new Date(),
      updatedAt: new Date(),
    } as T;
    this.store.set(entity.id, entity);
    return entity;
  }

  async delete(id: string): Promise<boolean> {
    return this.store.delete(id);
  }
}
```

2.3 SQL – Complex Analytical Query

```
-- PostgreSQL 15+: monthly revenue cohort analysis
WITH monthly_orders AS (
    SELECT
        DATE_TRUNC('month', o.created_at)      AS order_month,
        DATE_TRUNC('month', u.created_at)      AS cohort_month,
        u.id                                   AS user_id,
        SUM(oi.quantity * oi.unit_price)       AS revenue
    FROM orders o
    JOIN order_items oi ON oi.order_id = o.id
    JOIN users u        ON u.id = o.user_id
    WHERE o.status = 'completed'
      AND o.created_at >= '2023-01-01'
    GROUP BY 1, 2, 3
),
cohort_size AS (
```

```
    SELECT cohort_month, COUNT(DISTINCT user_id) AS size
    FROM monthly_orders
    WHERE order_month = cohort_month
    GROUP BY 1
),
retention AS (
    SELECT
        mo.cohort_month,
        EXTRACT(MONTH FROM AGE(mo.order_month, mo.cohort_month)) AS months_since_join,
        COUNT(DISTINCT mo.user_id)  AS active_users,
        SUM(mo.revenue)             AS total_revenue
    FROM monthly_orders mo
    GROUP BY 1, 2
)
SELECT
    r.cohort_month,
    r.months_since_join,
    cs.size                                          AS cohort_size,
    r.active_users,
    ROUND(r.active_users * 100.0 / cs.size, 2)       AS retention_pct,
    ROUND(r.total_revenue / NULLIF(r.active_users,0), 2) AS arpu
FROM retention r
JOIN cohort_size cs USING (cohort_month)
ORDER BY r.cohort_month, r.months_since_join;
```

2.4 Bash – Deployment Script

```
#!/usr/bin/env bash
set -euo pipefail
IFS=$'\n\t'

# ■■ Config ■■■
```

```
kubectl set image "deployment/$APP" "$APP=$IMAGE" --namespace "$ENV"
kubectl rollout status "deployment/$APP" --namespace "$ENV" --timeout=120s

log "Deployment complete: $IMAGE -> $ENV"
```

3. Tables

Tables vary in column count, alignment, and content complexity. Some cells contain **bold text**, *italic annotations*, or ***both***. The last table is intentionally wide to test overflow handling.

3.1 API Endpoint Reference

Method	Endpoint	Auth	Rate Limit	Description
GET	/api/v2/users	**Bearer**	1000/hr	List all users
POST	/api/v2/users	**Bearer**	100/hr	Create a user
GET	/api/v2/users/{id}	**Bearer**	5000/hr	Get user by ID
PATCH	/api/v2/users/{id}	**Bearer**	200/hr	Partial update
DELETE	/api/v2/users/{id}	**Bearer + MFA**	50/hr	Soft delete user
GET	/api/v2/products	*API Key*	10000/hr	Product catalogue
POST	/api/v2/orders	**Bearer**	500/hr	Place an order
GET	/api/v2/orders/{id}/status	**Bearer**	5000/hr	Poll order status
POST	/api/v2/webhooks	***Admin only***	10/hr	Register webhook
GET	/health	None	Unlimited	Health check

Table 1 — REST API endpoint reference with auth and rate-limit info.

3.2 Library Comparison Matrix

Library	Language	Stars	License	Async	Notes
FastAPI	Python	72k	MIT	**Yes**	OpenAPI auto-docs; excellent typing support
Express	JavaScript	63k	MIT	Callbacks/Promises	Mature ecosystem; minimal by default
Gin	Go	77k	MIT	*Goroutines*	Very fast; low memory overhead
Axum	Rust	18k	MIT	*Tokio*	Type-safe extractors; zero-cost abstractions
Rails	Ruby	55k	MIT	Limited	Convention over config; full-stack
Laravel	PHP	77k	MIT	*Swoole opt.*	Rich ORM; batteries included
Spring Boot	Java	72k	Apache 2	*WebFlux*	Enterprise-grade; verbose config

NestJS	TypeScript	66k	MIT	**Yes**	Angular-style DI; opinionated structure

Table 2 — Web framework comparison. **Bold green** = native async support.

3.3 Performance Benchmark Results

Results below are from a **3-minute sustained load test** at *2000 concurrent connections* on an AWS c6i.2xlarge instance. Latency figures are in milliseconds. ***Lower is better.***

Framework	RPS	p50 (ms)	p95 (ms)	p99 (ms)	Error %	RAM (MB)
Axum (Rust)	148,204	6.1	12.4	19.7	**0.00%**	24
Gin (Go)	132,877	7.5	14.1	22.3	**0.00%**	31
FastAPI (Python)	28,440	35.2	68.9	97.1	0.12%	148
NestJS (TS)	24,901	40.1	79.3	112.5	0.08%	197
Spring Boot	19,322	51.7	103.2	188.4	0.31%	412
Rails 7	4,811	208.5	501.2	798.1	**1.20%**	321

Table 3 — Load test results. ***Red figures*** indicate values outside SLA thresholds.

4. Mixed Formatting & Edge Cases

4.1 Bold, Italic, and Bold-Italic Prose

Normal text flows into **bold text** and then *italic text* and then ***bold-italic text*** and back to normal. This sentence tests mid-word transitions: un**expect**ed and semi*auto*matic and over***load***ed.

The entire paragraph can be bold. Converters must not collapse the bold span or merge it with adjacent non-bold paragraphs. **Multiple. Bold. Sentences. In. A. Row.**

*An entirely italic paragraph follows. It contains a nested **bold island** inside italic text, which should produce bold-italic output: **this is bold-italic**. Then the italic continues normally.*

4.2 Inline Code Inside Prose

Call the function `fetch_paginated()` with a `page_size` argument. The return type is `list[dict[str, Any]]`. Do **not** confuse `None` with `null`; they are different types. The environment variable `DATABASE_URL` **must** be set before calling `migrate()`.

4.3 Warning & Note Callout Blocks

NOTE: The following configuration is *environment-specific*. Values for **production** must be sourced from the secrets vault, **never** hard-coded in repository files.

WARNING: Running `DROP TABLE users` in production is ***irreversible***. Ensure you have a verified backup before proceeding. This action will trigger a **P0 incident** if executed without a change-management ticket.

4.4 Deeply Nested Checklist

Below is a checklist where some items have *sub-items* (indented). Converters should preserve the indent level as nested list items.

- [x] Backend services healthy
  - [x] API gateway responding
  - [ ] Auth service token refresh tested
  - [x] Database connections pooled
    - [ ] Read replicas reachable
    - [ ] Write throughput within limits
- [ ] Frontend assets deployed
  - [ ] CDN cache invalidated
  - [ ] Service worker updated
- [x] Monitoring configured
  - [x] Prometheus metrics scraping
  - [ ] Grafana dashboards imported

- [ ] PagerDuty alerts wired
  - [ ] On-call rotation set for week

5. Typography & Special Characters

5.1 Special Characters & Entities

Ampersand: A & B. Angle brackets: <tag> and </tag>. Quotes: "double" and 'single'. Em dash: mdash — ndash – hyphen-minus -. Ellipsis: waiting... vs proper: ... Copyright: (C) Registered: (R) Trademark: (TM). Mathematical: $x^2 + y^2 = z^2$. Chemical: H_2O and CO_2. Currency: $ 1,234.56 | 1.234,56 EUR | 1,23,456 INR.

5.2 Long Unbreakable Tokens

The following is a long identifier that may challenge line-wrapping: `com.example.platform.services.authentication.providers.oauth2.handlers.RefreshTokenExchangeHandler`. And a long URL: `https://api.example.com/v2/organisations/org_01HXYZ/members?include=roles,permissions&page_size=100&cursor=eyJpZCI6MTIzfQ`.

5.3 Mixed-Language Code Block

```
# docker-compose.yml
version: "3.9"
services:
  api:
    image: registry.example.com/api:${TAG:-latest}
    environment:
      DATABASE_URL: "postgres://user:${DB_PASS}@db:5432/app"
      REDIS_URL: "redis://:${REDIS_PASS}@cache:6379/0"
      SECRET_KEY: "${SECRET_KEY}"
    depends_on:
      db:
        condition: service_healthy
    deploy:
      replicas: 3
      update_config:
        parallelism: 1
        delay: 10s
        failure_action: rollback

  db:
    image: postgres:16-alpine
    environment:
      POSTGRES_PASSWORD: "${DB_PASS}"
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U postgres"]
      interval: 5s
      timeout: 5s
      retries: 5
    volumes:
      - pgdata:/var/lib/postgresql/data

  cache:
    image: redis:7-alpine
    command: redis-server --requirepass "${REDIS_PASS}"
```

```
volumes:
  pgdata:
```

5.4 Final Checklist — Converter Validation

After running the converter, verify each item below. **All items should be checked** before the output is considered ***production-ready***.

- [ ] Headings H1–H3 present and correctly levelled
- [ ] Bold text preserved in all paragraphs
- [ ] Italic text preserved in all paragraphs
- [ ] Bold-italic combinations preserved
- [ ] Checklist items render as GFM task list syntax
- [ ] Checked boxes render as [x] and unchecked as [ ]
- [ ] Code blocks use fenced ``` syntax with language hint
- [ ] Inline code uses backtick notation
- [ ] All four tables render with correct column count
- [ ] Table header row bold and visually distinct
- [ ] Alternating row background not lost (acceptable)
- [ ] Special characters not garbled (&, <, >, quotes)
- [ ] Superscripts and subscripts handled gracefully
- [ ] No extra blank lines inserted between list items
- [ ] No content dropped across page boundaries
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```
volumes:
  pgdata:
```

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